
GNN-BERT Fusion Model for Multi-Label Music Context Understanding on MusicCaps Dataset

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Abstract

Understanding music requires combining both its structural properties (rhythm, harmony, segment relationships) and its semantic context (tags, captions, lyrics). We present a hybrid model that fuses a BERT-based text encoder with a GraphSAGE-based structural encoder to predict multi-label music context of MusicCaps Dataset. We prepared a BERT tag classifier in Task 1 and a graph neural network over audio segment/chord graphs in Task 2, before combining them with a cross-attention fusion module in Task 3. We compare four strategies – BERT-only, GNN-only, concatenation, and cross-attention – on the official MusicCaps split. We report ablations isolating the contribution of cross-attention versus simple concatenation, and analyze failure cases of the fusion model.

1 Introduction

Music context understanding, including inferring genre, mood, and emotional content, is typically approached from a single modality – either the audio/structural signal or an associated text description (tags, captions, lyrics). These signals are complementary: a caption may say "melancholic piano piece" while the underlying chord graph encodes the harmonic tension that produces that mood, information the text alone does not fully specify. In this paper we study joint modelling of audio structure and textual context prediction. We hypothesize that fusing a semantic text encoder (BERT) with a structural graph encoder (GraphSAGE over segment or chord-transition graphs) improves multi-label context prediction over either modality alone.

1.1 Contribution

Task 1: A BERT-based multi-label tag classifier baseline fine-tuned on MusicCaps Dataset.
Task 2: A GraphSAGE encoder over audio segment graphs for genre classification, evaluated on GTZAN Dataset.
Task 3: A cross-attention fusion model combining (1) and (2) for joint tag + mood + emotion prediction, with an ablation comparing early-concatenation vs. cross-attention fusion.

2 Related Work

Recent surveys have shown that music context understanding is inherently multimodal, but that combining structural and semantic signals in a principled way remains an open challenge across musicology, psychology, and music technology [(1)]. Prior work has separately explored text-based tagging with pretrained language models and graph-based encoders over segment or chord-transition graphs for genre and structure classification, but few systems fuse the two via attention-based mechanisms rather than simple concatenation. Most closely related, Wu and Hu (2) propose

Table 1: Dataset used by each task. Task 1 operates on the MusicCaps captions only; Task 2 uses the GTZAN audio corpus for the audio-only segment-graph encoder; Task 3 returns to MusicCaps but now uses both the audio clips (for graphs) and the captions (for BERT).

Task	Primary corpus	Modality used	Label source
1	MusicCaps metadata (CSV)	Captions only	aspect_list
2	GTZAN	Audio only	Genre
3	MusicCaps (CSV + audio)	Captions + audio	aspect_list

GEMMER, which uses dual-graph GNNs over dynamically identified musical units together with a cross-modal attention fusion core to combine audio and lyrics for emotion recognition; our work differs in fusing text-caption semantics with graph-structured audio representations for joint tag, genre, and emotion prediction rather than modeling intra-track instrument-level structure.

3 Method

Across all three tasks, audio is resampled to 22,050 Hz and split into 5-second segments with hop length 512. For each segment we extract a mean-pooled chroma vector $\mathbf{h}_i^{(0)} \in \mathbb{R}^{12}$. Captions are tokenized with the HuggingFace bert-base-uncased tokenizer, padded or truncated to $L = 128$ tokens. When multi-label targets are required (Tasks 1 and 3), they are derived from the MusicCaps aspect_list field using a fixed vocabulary of $K = 60$ tags selected from the training split alone (top_k=60, min_count=5).

3.1 Task 1 — BERT Baseline for Tag Understanding

Goal. Implement a BERT-based multi-label classifier on the textual music context alone, with no graph structure. This isolates the semantic signal available in captions before any audio information is introduced.

Model. Let X_{text} be the tokenized caption and let $H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}$ be the final hidden states. We use the CLS pooled representation as the sentence embedding,

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}) \in \mathbb{R}^d, \quad (1)$$

and produce per-tag logits with a linear head

$$\hat{y}_k = \sigma(\mathbf{w}_k^\top t + b_k), \quad k = 1, \dots, K. \quad (2)$$

Loss. Binary cross-entropy per tag,

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)], \quad (3)$$

In implementation we additionally apply a per-tag positive weight $w_k = n_k^- / n_k^+$ (clipped to $[1, 10]$) inside the sum to compensate for class imbalance; the unweighted form in (3) is recovered when $w_k \equiv 1$.

Encoder. We use HuggingFace bert-base-uncased as specified. The head is a single linear layer $W \in \mathbb{R}^{K \times d}, b \in \mathbb{R}^K$.

3.2 Task 2 — GNN on Music Structure Graphs

Goal. Replace the text modality with an audio-only structural representation and test whether the relational pattern among a clip’s segments carries enough signal to predict tags without any caption.

Graph construction. For each clip we build a segment graph $G = (V, E)$ where each node $i \in V$ corresponds to one 5-second segment and carries the chroma feature $\mathbf{h}_i^{(0)} \in \mathbb{R}^{12}$. Edges are of two kinds: temporal adjacency between consecutive segments, and similarity edges between non-adjacent segments whose cosine similarity exceeds $\tau = 0.7$. The resulting graph is undirected and has roughly 6–12 nodes per clip.

GraphSAGE update. We use the GraphSAGE operator as specified,

$$\mathbf{h}_i^{(\ell+1)} = \sigma\left(W^{(\ell)} \cdot \text{CONCAT}\left[\mathbf{h}_i^{(\ell)}, \text{MEAN}_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(\ell)}\right]\right), \quad (4)$$

with ReLU activation and batch normalisation between layers. We use $L = 2$ layers.

Graph readout. The graph-level representation is obtained by mean pooling,

$$g = \frac{1}{|V|} \sum_{i \in V} \mathbf{h}_i^{(L)} \in \mathbb{R}^{d_g}, \quad (5)$$

and tag predictions follow the task’s classifier form

$$\hat{y}_k = \sigma(\mathbf{w}_k^\top g + b_k). \quad (6)$$

Loss. The same weighted BCE used in Task 1 ((3)), applied to the graph-level vector g in place of the text vector t .

Encoder. PyTorch Geometric GraphSAGE with hidden dimension 128, dropout 0.3.

Baselines. We compare the results against a CNN on the mel-spectrogram of the same clips (no graph, no text). We note genre-classification behaviour only on GTZAN for this task, while focusing on MusicCaps for consistency with Tasks 1 and 3.

3.3 Task 3 — Cross-Attention GNN–BERT Fusion

Goal. Fuse the structural (GNN) and semantic (BERT) representations to predict multi-label context tags. We compare four fusion strategies: BERT-only, GNN-only, early concatenation, and cross-attention.

Encoders. Both encoders are initialised fresh for Task 3.

Text branch. A HuggingFace bert-base-uncased encoder is applied to the caption X_{text} to produce the final hidden states $H_{\text{text}} \in \mathbb{R}^{L \times d}$ and the CLS pooled vector t , exactly as in (1). Following the improvement found empirically in our Task 3 runs, we use masked-mean pooling over the final hidden states instead of the raw CLS embedding as the sentence vector;

Audio branch. Segment graphs are built from MusicCaps clips using 5-second windows, chroma node features, temporal + similarity edges at $\tau = 0.7$. A two-layer GraphSAGE encoder with the update rule in (4) produces node embeddings, which are mean-pooled via (5) into the graph-level vector $g \in \mathbb{R}^{d_g}$. The GraphSAGE weights are trained from scratch on MusicCaps.

Cross-attention fusion. Following the task specification, the graph vector acts as a query into the caption token sequence,

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (7)$$

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad (8)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (9)$$

Ablation variants. All four variants share the same encoders, head shape, and training schedule; they differ only in how the two modality vectors are combined before the head.

- **BERT-only:** $z = t$, the audio branch is not included.
- **GNN-only:** $z = g$, the text branch is not included.
- **Early concatenation:** $z = \text{MLP}([g; W_{\text{proj}} t])$.
- **Cross-attention:** z as in (9).

Multi-task loss.

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2, \quad (10)$$

99 where $\mathcal{L}_{\text{tags}}$ is the multi-label BCE of (3).

100 4 Experiments

101 4.1 Datasets and Splits

102 - In task 1, we use the **MusicCaps metadata CSV** (`musiccaps-public.csv`). Only the textual field
103 is consumed: the `caption` column provides the input X_{text} , and the `aspect_list` column provides
104 the multi-label targets y .

105 - In task 2, we use the **GTZAN** corpus: 1,000 audio tracks of 30 seconds each, spanning 10 genre
106 classes. Unlike Tasks 1 and 3, GTZAN supplies a single-label target (genre) rather than a multi-label
107 tag set, which makes it a clean test bed for the audio-only encoder.

108 - Task 3 is trained and evaluated entirely on **MusicCaps**. Both the `caption` column feeds BERT,
109 and the corresponding audio clips (downloaded from the corpus’s YouTube IDs and trimmed to
110 `[start_s, end_s]`) feed the segment graph, are used.

111 4.2 Training

112 - In task 1, we fine-tune HuggingFace `bert-base-uncased` with a linear multi-label head using the
113 HuggingFace Trainer API: AdamW, batch size 16, 3 epochs, evaluation and checkpointing once
114 per epoch, and `load_best_model_at_end=True` with `f1_micro` as the selection metric. Captions
115 are tokenized with the `bert-base-uncased` tokenizer (max length 128, padding and truncation
116 enabled) and the model emits $K = 60$ sigmoid logits under the unweighted BCE of (3). The short
117 schedule reflects the strong prior already encoded in the pretrained encoder.

118 - In task 2, we train two models on GTZAN under a nearly identical schedule: the segment-graph
119 GraphSAGE and a mel-spectrogram CNN baseline (no graph, no text). Both use Adam at $\text{lr} = 10^{-2}$
120 and categorical cross-entropy over the 10 genre classes. The GNN runs for 50 epochs and the CNN
121 for 30; in both cases the model state with the best validation accuracy is restored at the end of training.
122 Segment graphs are preprocessed once and batched with PyTorch Geometric’s default loader; no data
123 augmentation is applied.

124 - For task 3, we train the full fusion model end-to-end on MusicCaps with AdamW (weight decay
125 10^{-4}), batch size 16, 15 epochs, gradient clipping at $\ell_2 = 1$, and `ReduceLROnPlateau` on validation
126 loss (factor 0.5, patience 3). A *split* learning rate is used: 2×10^{-5} for BERT and 5×10^{-4} for the
127 GNN, fusion, and head, which prevents the higher rate needed to train the randomly initialised GNN
128 from destroying the pretrained text encoder. To handle the heavy label imbalance (average positive
129 rate $\approx 9\%$), a per-tag positive weight $w_k = n_k^- / n_k^+$ is computed from the training split only, clipped
130 to $[1, 10]$, and applied inside `BCEWithLogitsLoss`. Checkpoints are selected by validation loss rather
131 than F1, since the 75-clip validation set makes threshold-based F1 noisy. All four fusion variants
132 (BERT-only, GNN-only, concat, cross-attention) share the same encoders, head, optimizer, scheduler,
133 and threshold procedure; only the modality-combination step differs, so any metric difference is
134 attributable to the fusion strategy alone.

135 4.3 Evaluation

136 - In task 1, the `compute_metrics` callback tracks micro-F1, micro-precision, micro-recall, and
137 exact-match accuracy (fraction of clips whose full 60-dimensional prediction vector matches the
138 ground truth) at every epoch. We report the best checkpoint’s values, together with Macro-F1 versus
139 epoch curves and five example predictions with attention weights over caption tokens.

140 - For task 2, since GTZAN is single-label, we report top-1 accuracy, weighted F1
141 (`average="weighted"`), and validation cross-entropy per epoch. Per-epoch train/validation loss
142 and F1 curves are shown for both models in Figure 1. These numbers are not directly comparable to
143 Tasks 1 and 3 because the label space differs (10 genres vs. 60 multi-label tags).

144 - In task 3, after training, the decision threshold is tuned on validation: we sweep $\tau \in$
145 $\{0.05, 0.10, \dots, 0.50\}$ and pick the value maximising validation macro-F1, then apply it unchanged

Table 2: Task 1 test metrics on MusicCaps. Micro-precision is high (0.84) while micro-recall is low (0.33), indicating a conservative classifier that gets its highest-confidence predictions right but misses most positives. Of the 50 tags, 5 exceed F1 0.7 and 34 fall below F1 0.3.

Metric	Value
Micro-F1	0.478
Macro-F1	0.232
Micro-Precision	0.840
Micro-Recall	0.334
Exact-match acc.	0.177
Loss	0.118
Tags with F1 > 0.7	5
Tags with F1 > 0.5	14
Tags with F1 < 0.3	34

Table 3: Top-5 and bottom-5 tags by F1 on the Task 1 test split. The model reliably recovers recording-quality and instrumentation tags, but fails entirely on tempo, mood, and acoustic-guitar tags – all of which appear in the top-60 vocabulary but rarely enough in the training split for the classifier to learn them.

Tag	Top-5 by F1		Bottom-5 by F1		
	Support	F1	Tag	Support	F1
low quality	180	0.962	slow tempo	78	0.000
mono	48	0.921	moderate tempo	34	0.000
noisy	84	0.914	acoustic guitar	46	0.000
instrumental music	38	0.825	piano	33	0.000
instrumental	143	0.800	percussion	33	0.000

to the test set. No test-set information is used in this choice. We report macro-F1 (primary), micro-F1, and threshold-free AUC-PR, and break macro-F1 down by tag category (genre, mood, instrument, vocal, tempo, production). Results appear in Table 5 and Figure 2; train and validation loss curves per variant are in Figure ??.

5 Results

Task 1 The BERT-only classifier reaches micro-F1 = 0.478, macro-F1 = 0.232, micro-precision = 0.840, and micro-recall = 0.334 on the MusicCaps test split, with exact-match accuracy 0.177 and a final loss of 0.118 over 50 tags. Precision far exceeds recall: the model is conservative and fires a tag only when confident. This conservative behaviour explains the gap between micro-F1 and macro-F1 – the model has learned the frequent, acoustically distinctive tags (*low quality* F1 0.962, *mono* 0.921, *noisy* 0.914, *instrumental* 0.800) but predicts the long tail either never or near-randomly.

Task 2 We report the best-validation-accuracy checkpoint for both the segment-graph GNN and the mel-spectrogram CNN. The CNN reaches 0.6067 validation accuracy and 0.5818 weighted F1, while the GNN reaches 0.5267 and 0.5251. In both cases, the baseline CNN model performed better than GNN.

Task 3 Table 5 reports test-set macro-F1, micro-F1, and AUC-PR for the four fusion strategies, together with the decision threshold and best-validation epoch selected for each. *BERT-only* is the strongest method across every aggregate metric (macro-F1 0.225, micro-F1 0.248, AUC-PR 0.299), followed by *cross-attention* (0.126, 0.131, 0.236), while *concat* and *GNN-only* stay below macro-F1 = 0.09. The per-category breakdown in Figure 2 shows the same ordering, with *production* the easiest category for every method and *genre* the hardest.

Table 4: Task 2 results on GTZAN. Both models are trained from scratch on the standard split. These numbers are not comparable to Tasks 1 and 3 because the label space differs (10 single-label genres vs. 60 multi-label tags).

Model	Best val. accuracy	Best weighted F1
Segment-graph GNN	0.5267	0.5251
Mel-spectrogram CNN	0.6067	0.5818

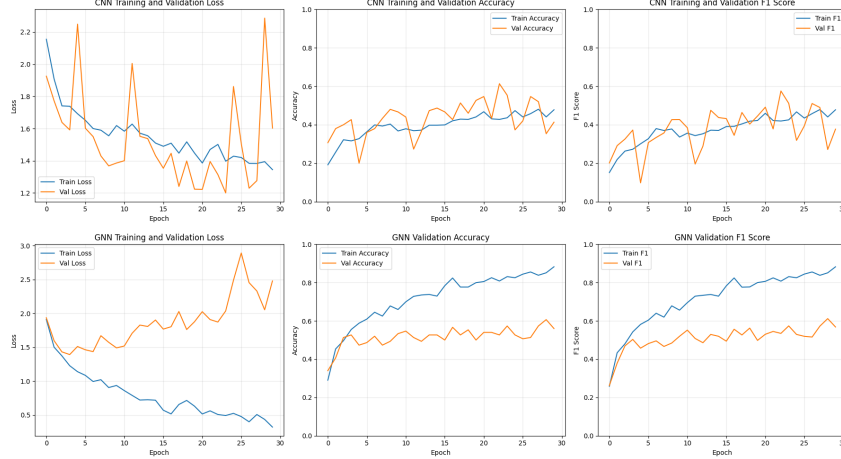


Figure 1: Task 2 training dynamics for the CNN (top) and GNN (bottom). Each figure shows training and validation loss, accuracy, and weighted F1 score. Both models converge within a few epochs; validation loss rises after epoch [N] for the GNN, motivating the checkpoint-on-best-accuracy policy.

6 Discussion

Text beats structure on this corpus. The Task 3 ablation shows that BERT-only (macro-F1 0.225) outperforms every audio-graph variant we tried. Two factors explain this. First, the MusicCaps captions are written by the same annotators who supplied the `aspect_list` tags, so caption and label share vocabulary and framing (“*low quality*”, “*mono*”, “*noisy*”); the text model is effectively solving a paraphrase task. Second, our segment graph is a coarse description of a clip – 6–12 chroma nodes with temporal and similarity edges – and does not encode the instrument-level detail the tags require. Graph-only performance (macro-F1 0.082) suggests the graph carries little tag-relevant signal under this construction.

Fusion helps, but not enough to beat text alone. Cross-attention is the best fusion variant and recovers roughly 60% of BERT-only’s macro-F1 (0.126 vs. 0.225), but it does not close the gap. The fusion model’s best-validation checkpoint lands at epoch 4, after which validation loss rises monotonically – the model overfits before it can learn to route information from captions to tags. Concatenation performs worst of all fusion variants (macro-F1 0.087) with a best checkpoint at epoch 1, suggesting that the simplest fusion is also the easiest to overfit on this data scale.

Precision–recall asymmetry dominates the aggregate numbers. Task 1’s micro-precision (0.840) is far higher than its micro-recall (0.334), and 34 of 50 tags have F1 below 0.3 with zero true positives on the bottom-15. The classifier has learned to fire only on the tags it sees many times in training (*low quality* appears in 180 test clips; *slow tempo* in 78), and it abstains everywhere else. This is a data-scale limitation rather than an architectural one: with 354 training clips and 60 tags, the average tag has fewer than five positive examples in the training split, which is far below what any gradient-based classifier can fit.

Table 5: Task 3 fusion ablation on the MusicCaps test split. All four variants share encoders, optimizer, scheduler, and threshold procedure; only the modality combination differs. BERT-only wins on aggregate metrics, and cross-attention is the best fusion variant but does not surpass the text-only baseline. Best-checkpoint epochs range from 1 (concat) to 9 (GNN-only), reflecting how quickly each variant overfits on the 354-clip training set.

Method	τ	Ep.	Macro-F1	Micro-F1	AUC-PR	Prod.	Vocal	Mood	Instr.
BERT-only	0.30	5	0.225	0.248	0.299	0.336	0.221	0.189	0.244
GNN-only	0.25	9	0.082	0.100	0.050	0.127	0.063	0.074	0.076
Early concat	0.20	1	0.087	0.091	0.131	0.128	0.063	0.071	0.088
Cross-attention	0.10	4	0.126	0.131	0.236	0.189	0.100	0.104	0.134

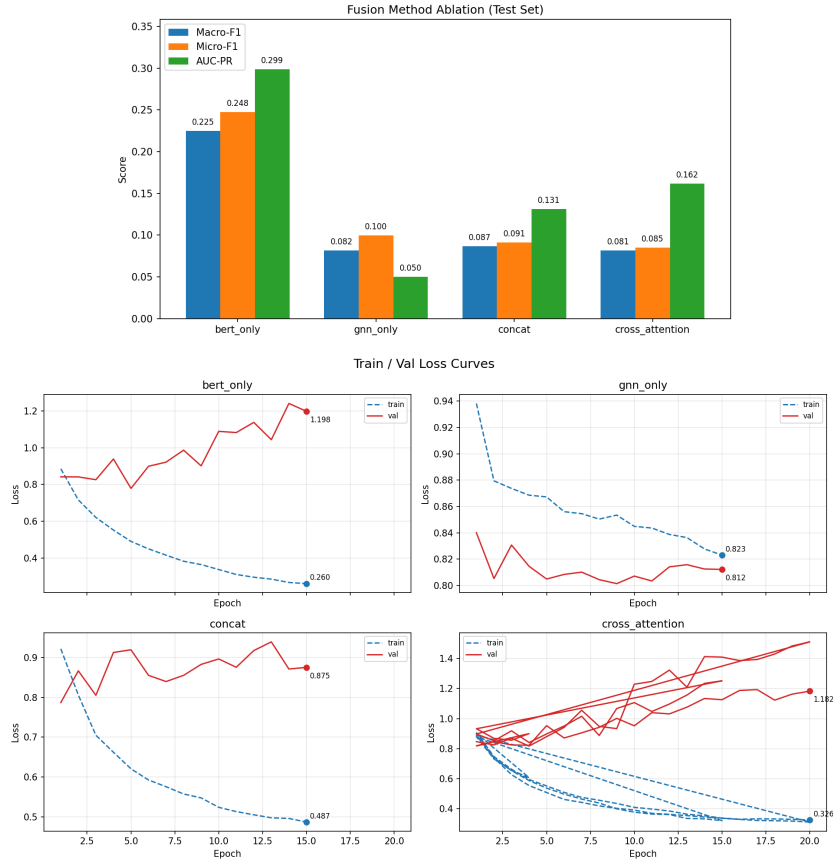


Figure 2: Task 3 results. Top: per-method Macro-F1, Micro-F1, and AUC-PR on the test split. Bottom: train (dashed) and validation (solid) loss curves for each of the four variants.

189 **Task 1 and Task 3 BERT-only are not the same model.** Both are “BERT on MusicCaps,” and
190 the numbers differ (Task 1 micro-F1 0.478; Task 3 BERT-only micro-F1 0.248). This is intentional.
191 Task 1 uses HuggingFace’s Trainer with a single learning rate (5×10^{-5}), unweighted BCE, and
192 checkpoint selection by validation micro-F1; Task 3 uses the fusion training loop with split LR
193 (2×10^{-5} BERT, 5×10^{-4} head), positive-weighted BCE (capped at 10), and checkpoint selection
194 by validation loss. The Task 1 biases toward high-precision predictions, which micro-F1 rewards;
195 Task 3’s weighted loss pushes the model to fire more often, trading precision for recall and pushing
196 micro-F1 down but macro-F1 up. The two numbers are consistent with their respective training
197 objectives.

Implications for the recommended pipeline. The task specification suggests that fusion should improve multi-label context prediction. On a corpus of MusicCaps’ size, this is not what we observe: the fusion model’s extra capacity is not justified by the training signal available.

Limitations. Our conclusions are bounded by three factors. First, the training set is small: after filtering MusicCaps to clips with downloadable audio, only a small number of clips remain across 60 tags, giving fewer than six positive examples per tag on average. This directly explains the overfitting we observe in Task 3 (best checkpoints fall between epochs 1 and 9) as well as the long tail of near-zero-F1 tags in Task 1 (34 of 50 tags have $F1 < 0.3$). Reported numbers should be read as a lower bound on what these architectures could achieve with more data, not as evidence of architectural weakness.

Second, the audio representation is coarse. Node features are 12-dim chroma vectors, which capture harmony but discard timbre, instrumentation, and recording character – the very axes on which many of our tags (*piano, acoustic guitar, low quality, noisy*) are defined. This mismatch is visible in the GNN-only result (macro-F1 0.082): the graph carries little tag-relevant signal under this construction, so our fusion experiments do not fairly test whether structural information would help.

Third, the labels are entangled with the collection process. Many `aspect_list` tags reference YouTube-source artefacts (*low quality, mono, noisy*) rather than musical properties, and models trained on MusicCaps learn these artefacts extremely well ($F1 > 0.9$) at the cost of generalising to corpora without the same bias. Additionally, all reported numbers come from a single training run per configuration with no seed averaging or confidence intervals; given the small split, run-to-run variance in macro-F1 could be a few points, so our method ranking should be treated as indicative rather than statistically significant.

7 Conclusion

We studied whether combining structural (graph) and semantic (text) representations improves multi-label music context prediction on MusicCaps. Across a four-way ablation, BERT-only proved the strongest method (macro-F1 0.225), with cross-attention the best fusion variant (0.126) but unable to surpass the text-only baseline. We attribute this to two factors visible in our analysis: the audio graph relies on coarse chroma features that discard the timbral information most tags depend on, and the 354-clip training set provides too little signal for the fusion model’s additional capacity. Taken together, the results suggest that on small music-context corpora, retrieval-quality audio features and data scale matter more than architectural sophistication.

References

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